DATE: _____

Project Data Book

Smartsteth: A Convolutional Neural Network (CNN) Driven Auscultation Instrument for Early Detection of Lower Respiratory Tract Diseases Using Respiratory Sound Analysis for Rural Health Units

Teokisley Gazelle J. Jompella, Ahnycca Krish N. Malimban, and Kiesha Mekhaela B. Relatado

July - October 2025

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Be mindful while doing your research! DATE: _____

JULY 2025

(Before Experimentation)

July 12, 2025

The final selection of our research group members was a significant step in ensuring the success of our study. After weeks of observation and teacher consultations, the ones with a common interest and capability were chosen based on performance and commitment. This decision was made aligned with fairness and transparency, aiming to form a group that would perform efficiently and harmoniously. With clear roles and shared goal, we are ready to carry out our research with determination and unity.

July 11, 2025

the activity on the individual making of research titles was both a learning experience and a challenge. At first, we found it difficult to think of a research title that is original, relevant, and feasible. We struggled with choosing a topic that matched our interest while also meeting academic standards. However, after doing some reading and reflecting on current issues, we eventually came up with a title wherein we felt confident. This activity taught us the importance of critical thinking, patience, and creativity in the research process.

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Keep up the good work Kiddos!

DATE: _____

July 24, 2025

Despite class cancelations due to the storm we were still able to accomplish the Feasibility Funnel activity through a scheduled Google meet session. During the meeting, we evaluated each proposed title based on relevance, clarity, and feasibility. While the process helped us narrow down the most practical and achievable topics. it was emotionally challenging to set aside some topics that clearly reflected hard-work. There was a sense of guilt, especially when evaluating the ideas of our classmates. Nonetheless, the activity helped us understand the importance of being objective and realistic in research.

July 25, 2025

The submission of our revised research proposal marked another important step in our research journey. Although we successfully completed the task. the process was not easy especially since the majority of us had little to no experience in making a research proposal. We struggled with organizing our ideas, following correct format and making sure that every part was clear and complete. Thankfully, through constant communication, peer evaluation, and guidance from our teacher. We were able to revise and improve our work. This activity helped us gain a deeper understanding of the research process and develop skills that will be valuable in the next phase.

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Be specific on the questions!

DATE: _____

July 29, 2025

The activity on formulating descriptive and exploratory questions in groups was a productive and engaging experience. Working collaboratively allowed us to combine our ideas and perspectives, which made the process more meaningful. However we initially struggled with identifying the right type of question and aligning them with our chosen research title. This activity helped us understand the foundation of Research inquiry.

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keep it up kids.
you can do it!

DATE: _____

August 2025
(Before Experimentation)

August 4, 2025

We started by discussing what our research was truly about and why it mattered. It was challenging to narrow down the scope, but doing so helped us stay focused. we also clariefied what our would not corer to avoid confusion later on.

August 6, 2025

We reached out to our parents and school authorities to get formal approval for our project. Writing the rationale made us reflect deeply on why we chose this topic. The risk assesment helped us think ahead and plan for safety.

August 7, 2025

Visiting experts gave us a clearer picture of how our research would unfold. we took notes and photos to document. It was exciting to see our ideas starting to take shape.

August 8, 2025

On our second visit, we noticed details we missed the first time. This helped us improve our observation skills and refine our data collection plan. We felt more confident about the feasibility of our study.

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DATE: _____

August 12, 2025 — Filling out the ISEF forms was a bit tedious, but it taught us the importance of research ethics and proper documentation. We double checked everything to make sure it was complete. It felt good to meet official requirements.

August 13, 2025 — Writing chater 1 helped us organize our thoughts and clearly define our research goals. The research plan gave us a structured time-line to follow. It was satisfying to see our ideas laid out in a formal document.

August 14, 2025 — We gathered related studies and theories to support our topic, which deepened our understanding. creating the paradigm helped us visualize how our variables connect. this part made our research feel more grounded and academic.

August 15, 2025 — After reviewing everything, we submitted our work online-it was a proud moment for us as a group. we learned a lot throughout the process and felt accomplished.

Good luck on your grades!
don't loose track!

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label the components of your device!

you can do it!

DATE: _____

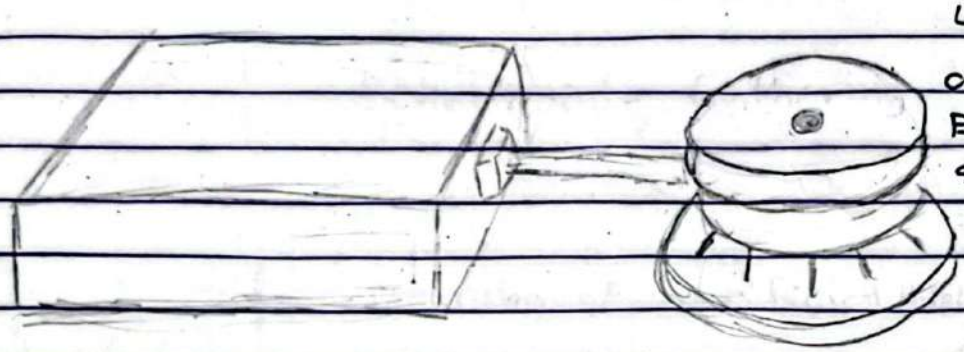
September 2025

(Start of Experimentation)

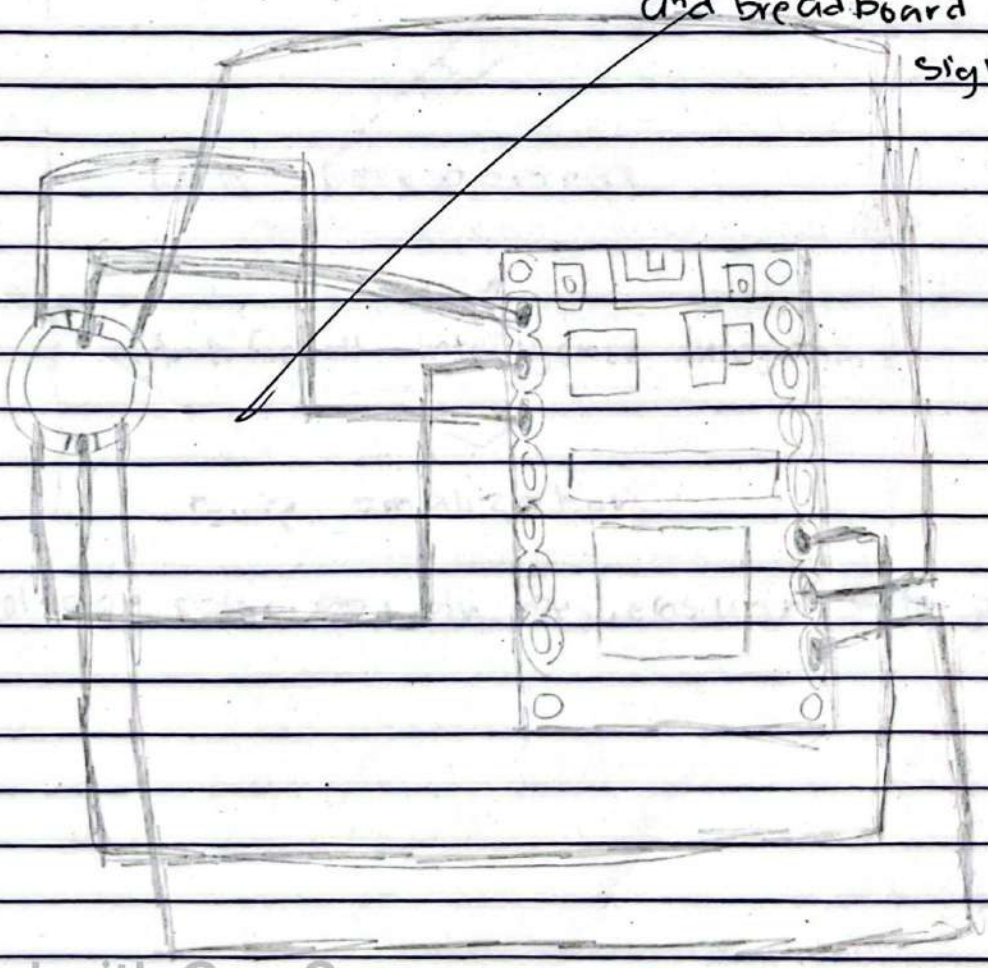
September 4, 2025

Visualizing smartsteth

↳ we designed a compact device combining ESP32, INM441, and a stethoscope for AI-powered diagnostics.



September 11, 2025 Cirkit Diagram → we mapped ESP32 connections using jumper wires and breadboard for smooth signal flow.



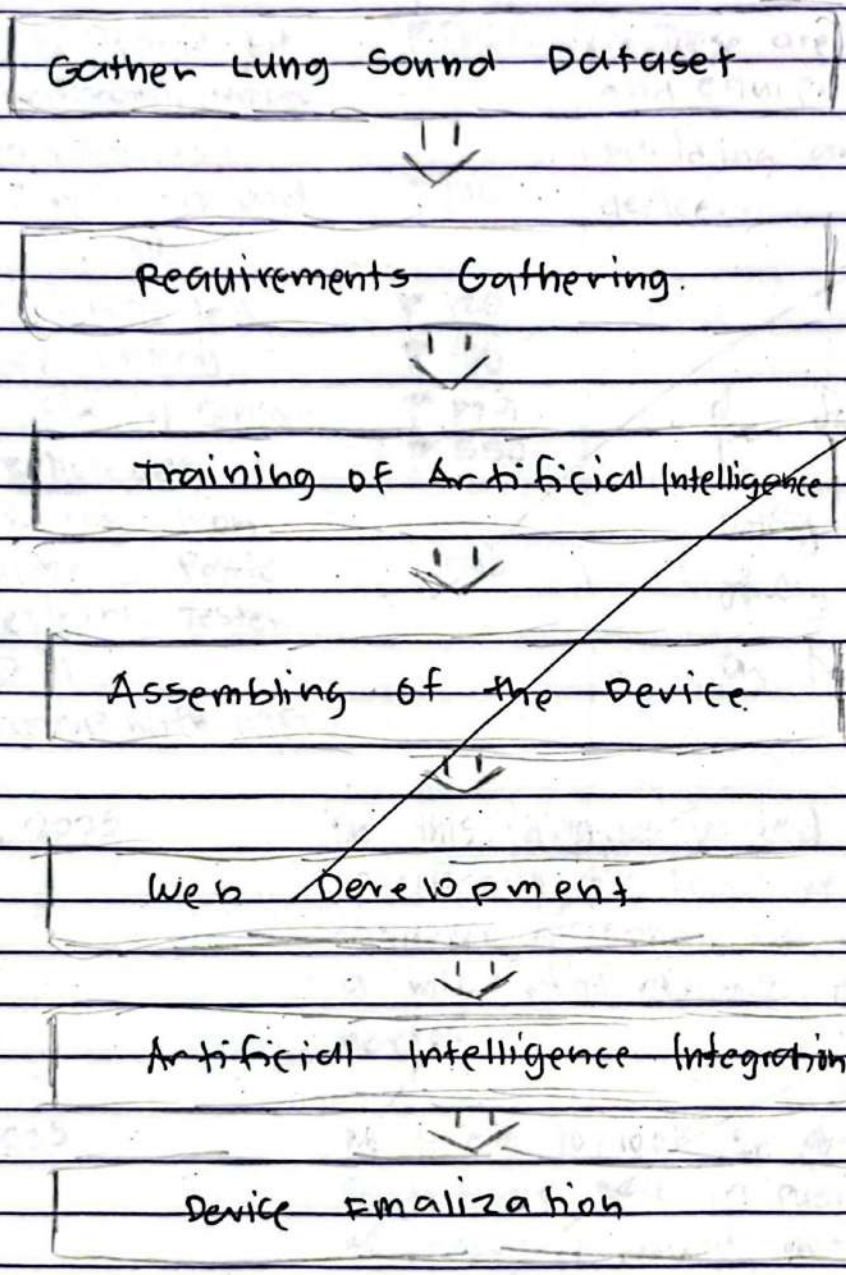
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Be careful and mindful!

DATE: _____

September 15, 2025

Schematic Diagram



It showed voltage paths and pin roles, ensuring safe and logical design.

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DATE: _____

September 19, 2025 Materials and Equipments needed

Materials	ESP32	₱409
	ESP32 Starter Kit (breadboard, jumper wires, sensors)	₱935
	Type C Upload cable	₱100
	Soldering led	₱120
	Box casing	₱106
	INM441 Sensor	₱179
	Stethoscope	₱550
Equipments	Soldering Iron	
	Soldering Paste	₱80
	Electronic Tester	
	Drill	
	Laptop with WIFI	

↳ These are the materials and equipments needed for building and testing our device.

~~The battle on the casting! Number your finding tools to be on the table!~~

September 24, 2025 On this day, we visited Fullbright College specifically Sir Kodic. We inquire a low frequency microphone but is given instead a mic that measure decibels and not hertz.

September 26, 2025 We tried to looks for condenser mic or electret mic here in puerto because if we order it would not arrive on time.

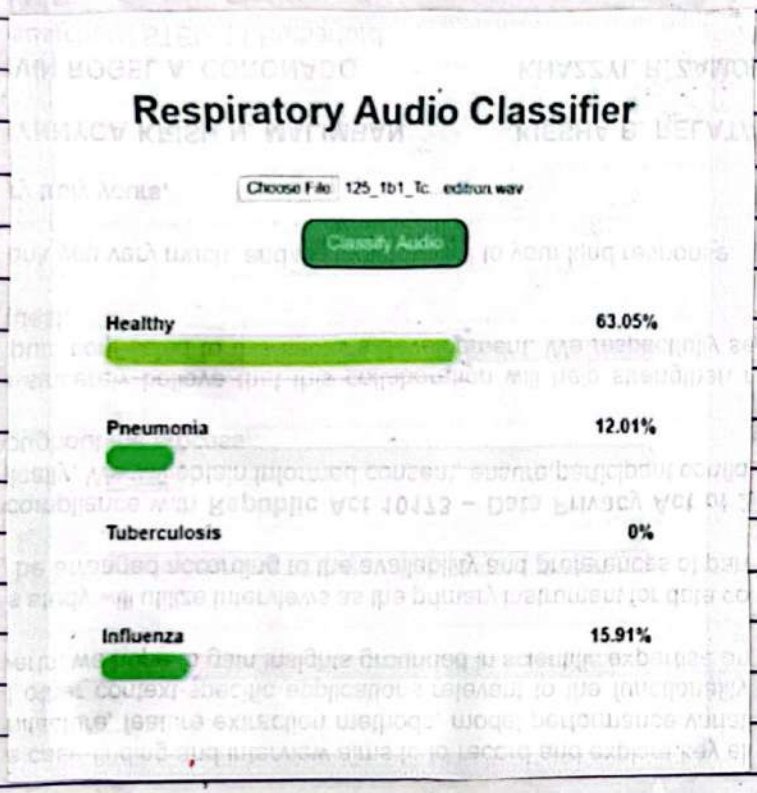
September 26, 2025 With SSTF nearing, we decided to ask advise from Sir Noche because our device wont work due to the lack of components and his verdict was to focus on the ai simulation for a much faster results.

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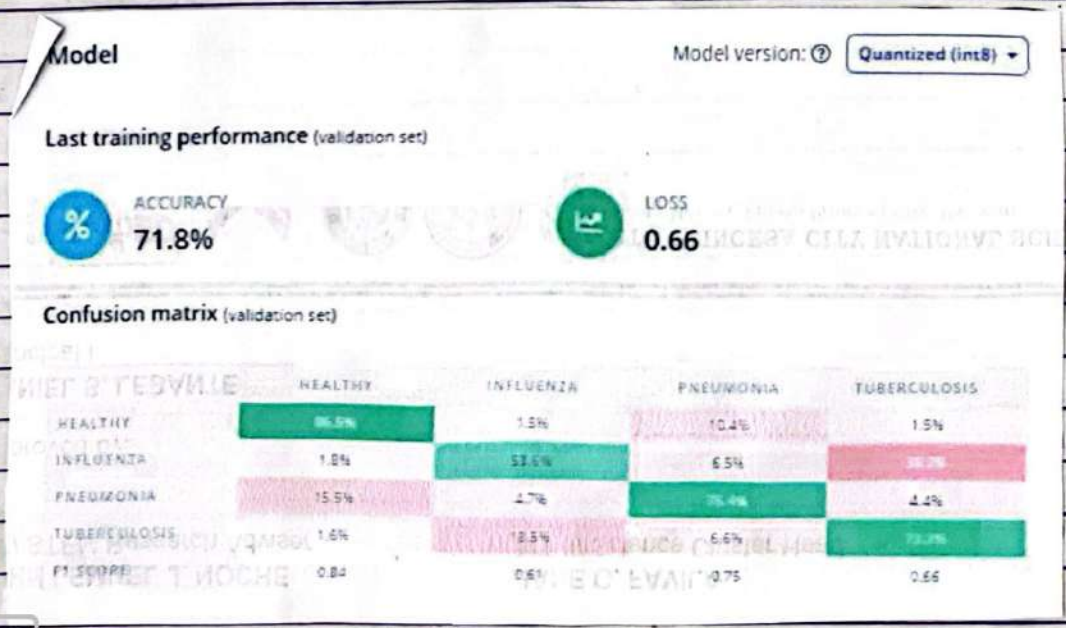
September 29, 2025
(Experimentation Period)

We find it hard to work individually so we went to Tejo Inn to finalize the paper and the simulation of our product.



↳ Initially we made a website to "pre-diagnose" or know what disease the audio recording have.

Great luck on your experimentation! Hoping for a successful results!



↳ This is the accuracy metric, this measures the accuracy of our ai simulation.

Love you guys!

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DATE:

September 30, 2025

Be prepared for the DSTF!

You Can do it!

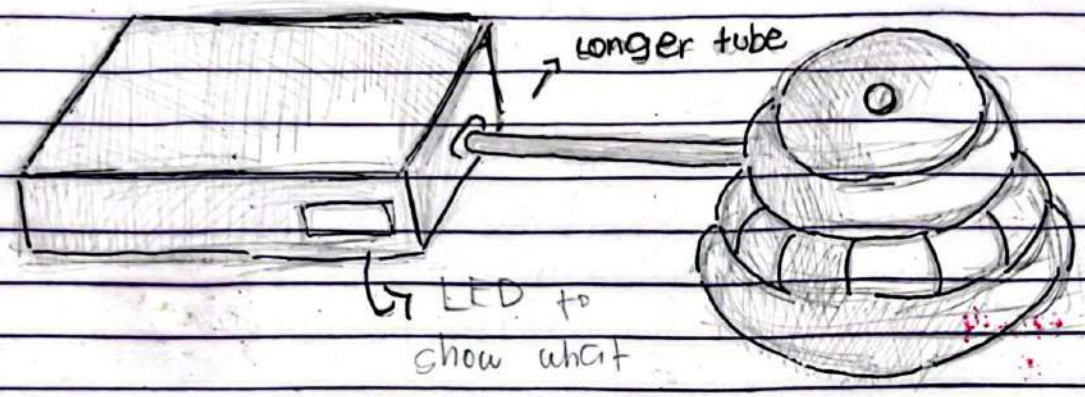
After learning we won our category in SSTF, we reflected on the challenges we faced and immediately ordered the components needed for DSTF. We revised our PP board and presentation, then in the morning we conducted an interview with a Nurse at Brgy. Sicsican Health center then we alloted our whole day in interviewing Head Nurses from 3 different satellites (Macarascas, Salvacion and San-Rafael), where we gathered insights for improving our device. After the components arrived we started to do the actual design/prototype of our device. We worked on the amplification of our mic, it is working but the audio was static.

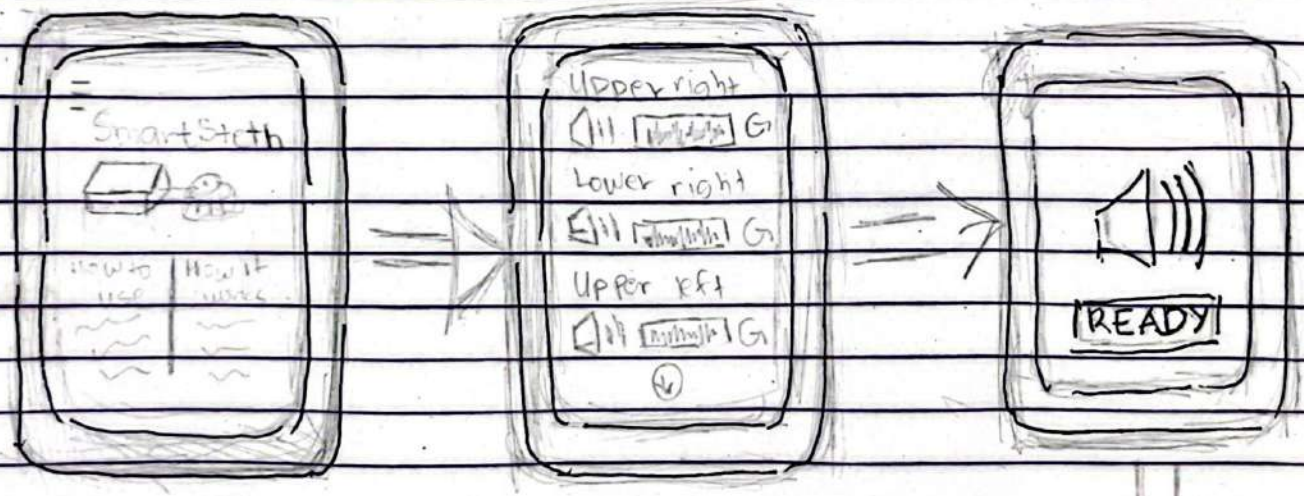
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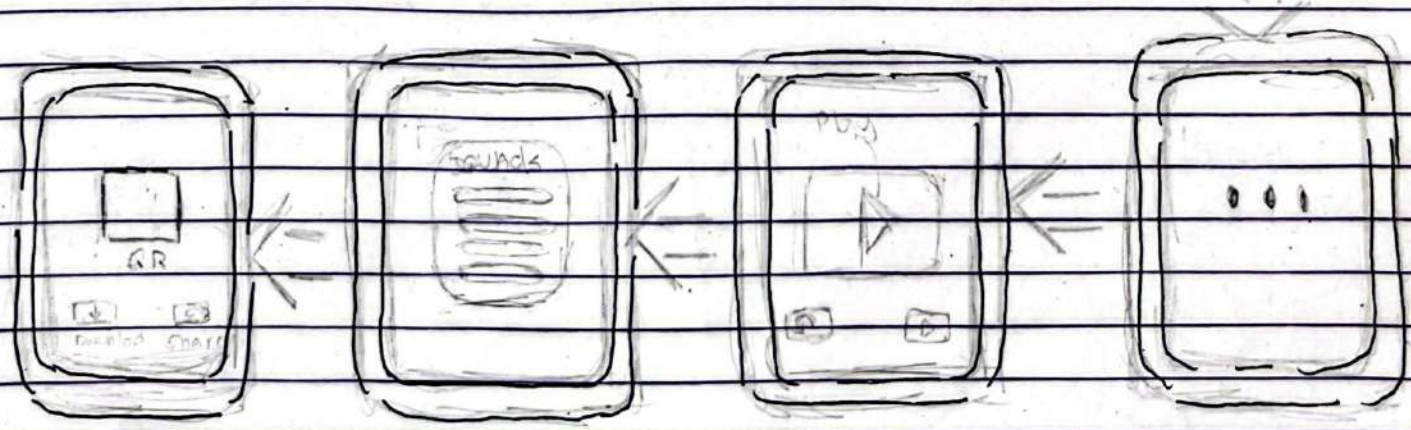
DATE: _____

October 2025

(End of experimentation)







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DATE: _____

Wait There's More!

Photo Documentations









We went to Salvacion, She answered our questions about user-need and preference. She also gave us free meds for she auscultated one of our members.



→ MACARASCAS



↳ San RAFAEL

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DATE: _____

October

(October 10, 2025) Haha it was not the end! We qualified for RSTF we were thrilled but exams was also coming for us so we decided to focus on one thing then the other. We applied for UM - Utility Model and also Patent Search in PITBI PSU campus. It was tiring but assuring that we may qualify for a patent search.

(October 28, 2025) We ordered the new components hoping we could integrate what the panelist want. However due to Typhoon Tino and Uwan deliveries were on hold :(

November

(November 10, 2025) Some arrived but the essentials did not, as our adviser was in Manila he said he will bring the components with him home unfortunately, the traffic was insane the rider couldn't catch Sir's flight so we opt to deliver it through LBC.

Same day, my lolo was going for a checkup a Pulmonologist! so I decided to come with him and observe how he was diagnosed then I asked the doctor if we could use his findings to validate our device. I love how experts are so supportive! he agreed :)

(November 12, 2025) Due to the typhoon Engr. Mark Tabalia was trapped in Cuyo, Palawan and cannot process our patent search. We went back to him this day and he conducted a training about doing patent search because he said it will be helpful for us as a researchers.

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DATE:

He made a commitment after we provide a letter to the president he will the department will issue a certificate that our device is novel and can passed IPOPHIL and may declare "on going" status.

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